Hippocampal-inspired memory in Minecraft

Does movement-conditioned visual memory learn spatial relationships that generalize across routes?

Completed pilot and offline activation audit

Qwen visual features, associative retrieval and candidate g/p populations

Research update | October 2026

Evidence: run 7246240. New controlled experiment remains pending.

Cognitive maps and associative memory

Behrens et al. (2018)

A cognitive map organizes relationships so an agent can make flexible inferences. Spatial navigation provides a concrete setting for studying this principle.

Whittington et al. (2022)

Recurrent movement-dependent position representations query past sensory experiences. The resulting model connects transformer retrieval with hippocampal models.

Our research question

After taking a different route, can the representation recover where the agent is and what it previously experienced there?

Sources: Behrens et al., Neuron (2018), doi:10.1016/j.neuron.2018.10.002; Whittington et al., arXiv:2112.04035

Our implementation

A self-supervised predictive memory. Qwen stays frozen.

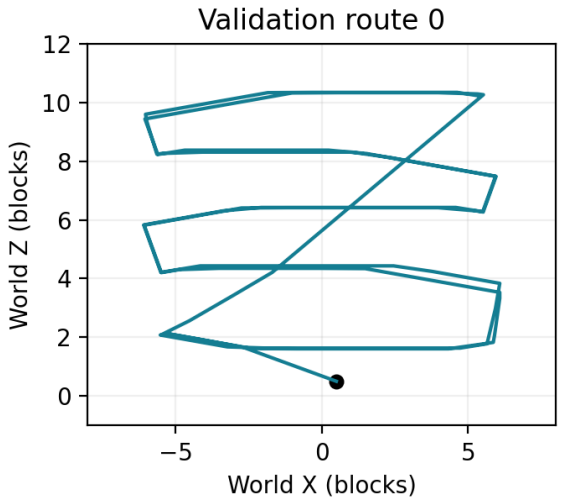
Component	Implementation
Visual input	Qwen2-VL-7B-Instruct vision features, projected to 64 dimensions
Movement input	Measured world-axis displacement and heading change
g state	32-dimensional GRU state updated from movement
Retrieval	Current g queries previous g keys and recalls past visual features
p state	32-dimensional tanh layer combining g and recalled vision
Learning objective	Predict current features before writing them into memory

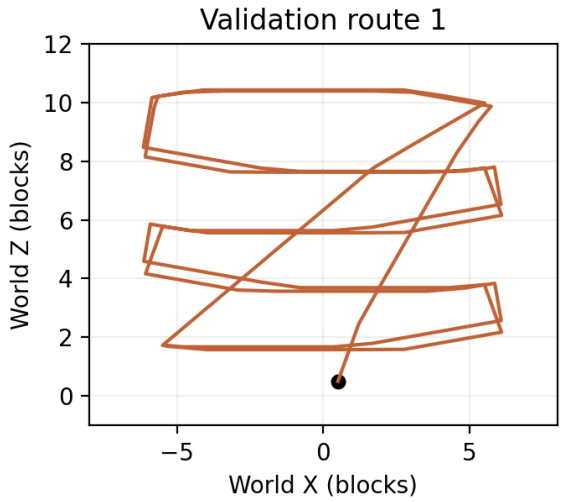
Our p layer is a learned conjunction. It differs from the original TEM outer-product construction. The g/p names denote candidate populations.

Implementation: src/mc_binding/hippocampal.py; src/mc_binding/hippocampal_study.py

Completed exploration pilot

Six geometries, two scripted routes each, 1,697 recorded frames, one model seed (731)





Training

Route 0 in four geometries.
100 epochs of visual prediction.

Evaluation

Route 1 in training geometries.
Both routes in one validation and one test geometry.

All 12 routes completed. Cached features align with recorded poses. Exploration and odometry use simulator telemetry. The memory does not receive absolute coordinates.

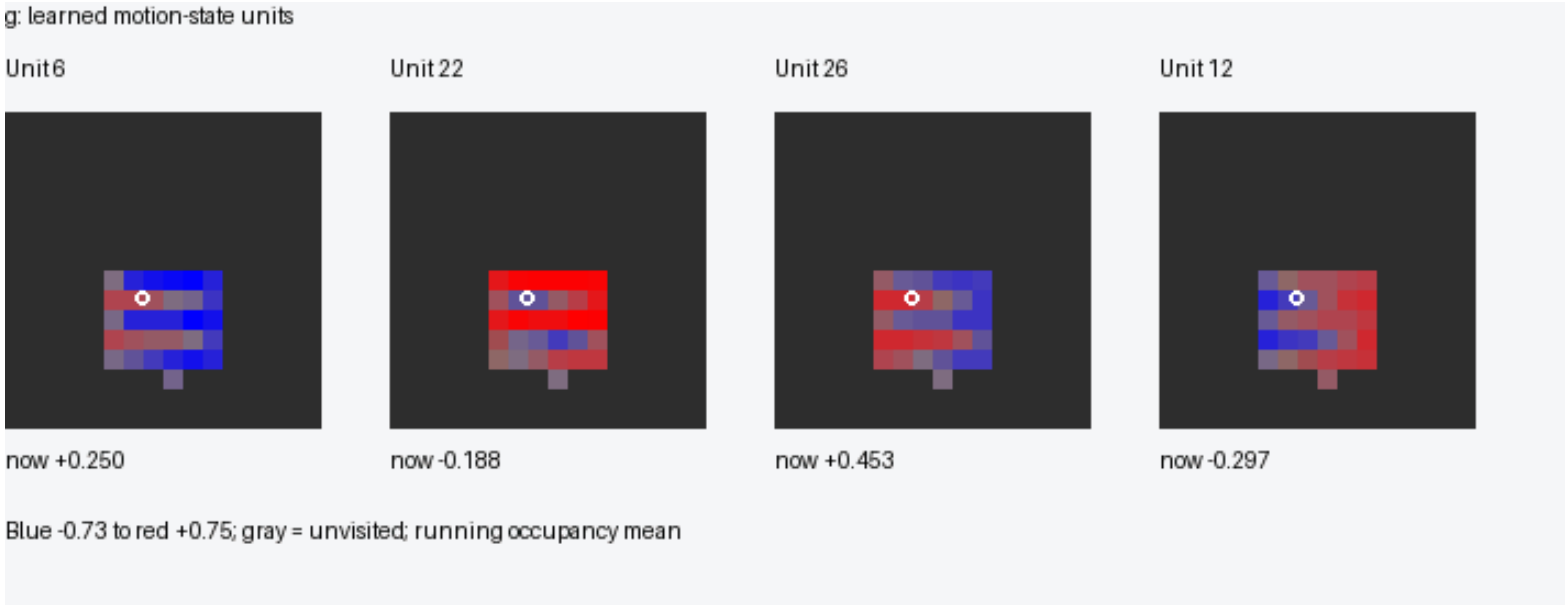
Source: sequences.json and manifest.json. Plots show recorded validation geometry f0013.

g activity during exploration

Four units selected by activation variance on training episodes



Validation route 0
Frame 125



Colors show accumulated mean activation in visited world bins. Gray cells are unvisited. The cursor marks the agent position.

GIF: assets/mapping-f0013-route0.gif

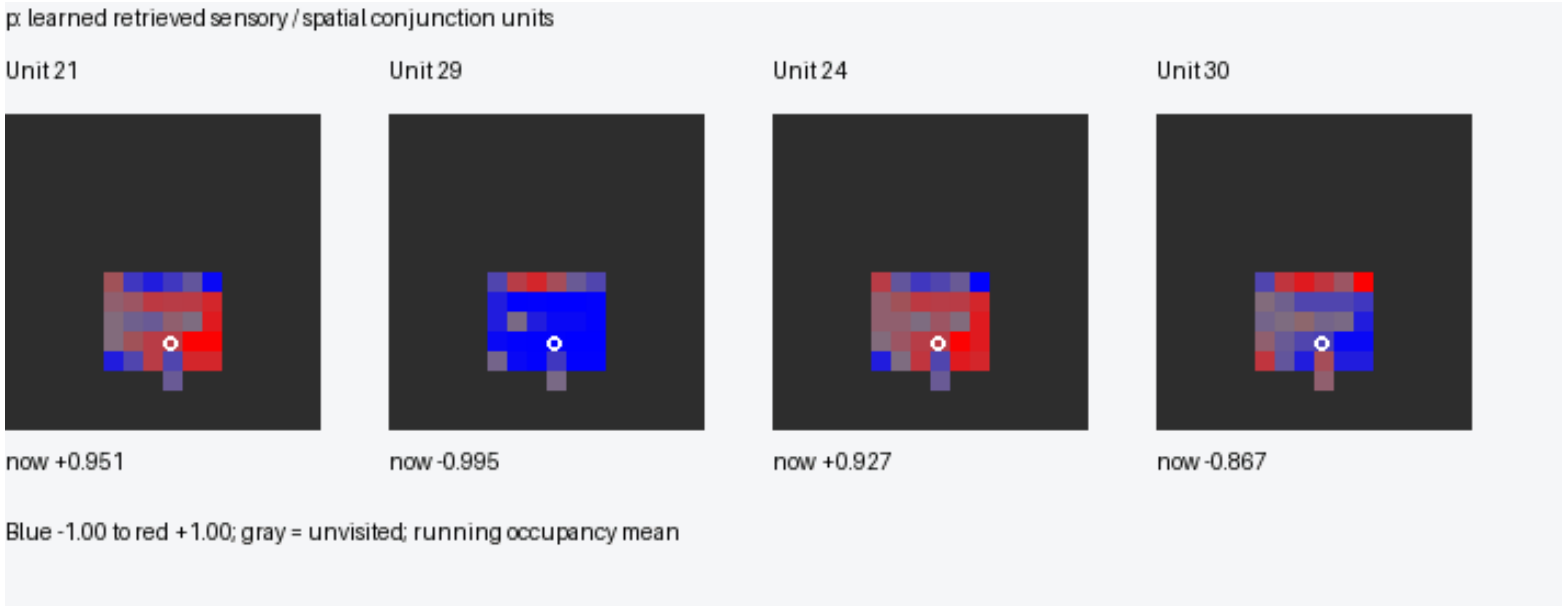
The plotting code supplies the fixed world axes. These panels do not establish grid-cell periodicity.

p activity combines motion state with recalled vision

Four units selected using the same training-only rule



Validation route 1
Frame 125



p depends on both movement history and retrieved sensory features. Its activity can change as the memory accumulates experience.

GIF: assets/mapping-f0013-route1.gif

Candidate conjunction units. Our implementation does not reproduce the original TEM p construction.

The camera view changes along a route

Recorded Minecraft frames, validation route 0



Frame 0



Frame 60



Frame 130

The same world can generate different visual features as heading and position change. A sensory-prediction objective may therefore preserve view information as well as location.

GIF: assets/mapping-f0013-route0.gif

PDF shows selected still frames. Open animations.html for the synchronized full activity replays.

Retrieval improves held-out sensory prediction

Mean normalized feature MSE across the two routes in the test geometry. Lower is better.



Replacing retrieval at inference increases prediction error. These controls test the trained pathway, rather than separately retrained architectures.

Source: analysis/cache-audit.json. Two correlated trajectories in one test geometry, not independent trials.

Activity repeats most strongly along the same route

Descriptive medians across unit correlations and episodes or geometries

Comparison	g	p
Same-route, same-heading matched revisits	0.955	0.718
Spatial maps across opposite routes	-0.181	-0.198

The two rows use different comparisons. The first pairs nearby positions with similar heading. The second correlates occupancy-averaged spatial maps across routes.

Route, heading and appearance change together. These results do not isolate the cause of the difference or rule out a joint position-and-heading representation.

Source: cache-audit.json. Frames and units are correlated. These medians are not replication counts or significance tests.

What the evidence supports

Useful associative retrieval

Ablating retrieval harms sensory prediction. The learned memory contributes to the completed task.

A movement-conditioned representation

g receives motion during inference. Visual prediction errors train its parameters, but observations do not directly correct g online.

Spatial generalization remains unresolved

Colored fields alone do not identify place/grid cells. Population-level spatial information may coexist with heading and route dependence.

Architecture fact: hippocampal.py. Experimental evidence: offline audit of model seed 731.

Proposed cognitive-mapping experiment

Implemented, but no new experimental results are available

Repeated locations with varied movement histories

Four intersecting loops repeatedly return to a shared anchor. Each return samples four cardinal headings.

Separate learning and evaluation routes

Train on two route orders. Evaluate reversed and reordered routes and held-out geometries.

Measure accessible spatial information

Fit evaluator-only location readouts on frozen training representations. Compare learned g/p with untrained states and a time/heading baseline.

The next run remains pending after an environment startup failure. No new cognitive-mapping results are available.

Design: docs/COGNITIVE_MAPPING.html. Location labels belong to the offline evaluator, not the memory objective.

Research conclusion and sources

We have evidence that movement-conditioned memory helps predict visual experience.

The next scientific question is whether its spatial relationships transfer across routes and headings.

Behrens et al. (2018). What Is a Cognitive Map? Organizing Knowledge for Flexible Behavior. doi.org/10.1016/j.neuron.2018.10.002

Whittington, Warren and Behrens (2022). Relating transformers to models and neural representations of the hippocampal formation. arxiv.org/abs/2112.04035

Local evidence: runs/oscar/hippocampal-mapping-7246240/analysis/cache-audit.json
Companion: animations.html (GIF paths appear beneath each replay)